

Universidade de Coimbra
Faculdade de Ciências e Tecnologia
Departamento de Engenharia Informática

Knowledge and Language 2026/2027

Masters in Informatics Engineering

Project

Professors:
Eliezer de Souza da Silva
eliezer.silva@uc.pt

Assistant:
Bruno Ferreira
brunof@dei.uc.pt

Attention: plagiarism will not be tolerated!

1 Goals

The main goal of the Knowledge and Language project is **to tackle a challenge** in the domain of the topics covered by the course. The project must include a meaningful **language / Natural Language Processing component** and at least one explicit **knowledge or uncertainty component**, such as:

- **structured or symbolic knowledge representation and reasoning** (e.g., logic, RDF, ontologies, Knowledge Graphs, Linked Data);
- **graph-based relational modelling** (e.g., Knowledge Graph embeddings, Graph Neural Networks, graph retrieval or graph reasoning);
- **probabilistic or uncertainty-aware modelling** (e.g., Bayesian models, calibration, sequential / structured probabilistic models, uncertainty-aware decision making).

The previous components should interact in a substantive way; simply using two unrelated tools is not sufficient. As long as their utilisation is well-supported, any computational tools can be used, as well as data sources, which can be on any domain.

2 Stages

The project should be developed by groups of **two** students, and must go through the following **stages**:

1. **Research & Problem Identification:** defining the challenge to tackle, supported by scientific literature. Two deliverables are expected, both of which can later be integrated in the **final report**:
    * **Literature Review:** a short report, with no more than **three pages in a 2-column format**[1], reviewing a small set of recent and directly relevant papers rather than presenting independent paper summaries. At least one paper should be from **2024–2026**, and the selected literature should preferably include work from strong peer-reviewed venues relevant to the

[1] Formatted according to the IJCAI guidelines, `https://www.ijcai.org/authors_kit`

topic (e.g., ICLR, ICML, NeurIPS, ACL, EMNLP, ISWC, ESWC, or another appropriate venue). The review should compare the problem settings, assumptions, data / knowledge sources, methods, evaluation protocols, main results, and limitations, and should clearly identify the gap or question that motivates the proposed project. Connections with the topics covered by the course must be explicit.

* **Project Proposal:** a single-page document describing the problem to tackle, setting the **goals** of the project and **dates of intermediate checkpoints**. The proposal must further enumerate, clearly, the **data** and **tools** to be used, proposed approach(es), at least one meaningful **baseline / comparison**, the intended **evaluation criteria or metrics**, and the expected contribution. It should also state the connections with the reviewed literature (e.g., initial inspiration, what will be made different) and with the topics covered by the course.

2. **Project Execution:** comprising the development of the project (design, implementation, evaluation), always **meeting the checkpoints** set and their discussion with the Professor.

3. **Final Report:** all the work developed must be described in a **final document**, in the form of a scientific paper, with no more than **eight 2-column pages**[2]. The following structure is suggested:

   (a) Introduction (including motivation, problem description, goal and approach; may be adapted from the Project Proposal);

   (b) Related Work (for which the Literature Review can be a starting point);

   (c) Data & Approach (an overview of the data / knowledge sources used, the approach adopted for tackling the problem, and the selected baselines / comparisons);

   (d) Implementation (including data exploitation, applied algorithms and tools used, and enough detail to make the experimental setup reproducible);

   (e) Experimentation (quantitative evaluation when possible, comparison with relevant baselines, examples of good and bad results, and appropriate error, uncertainty, sensitivity, or ablation analysis when relevant);

   (f) Conclusions (stressing the main achievements and main difficulties faced) and Future Work (what could still be done);

[2] Ideally, seven pages for content plus one for bibliographic references.

(g) Bibliographic References.

4. **Final Presentation:** in the end of the semester, there will be 10-minute (tentative) presentations of the projects developed, followed by a brief discussion between the Professor and colleagues (if present).

All documents can be **written either in Portuguese or English**.

3 Evaluation Criteria

The assessment scheme for the course is:

- **Research Work: 20% (4/20)** of the final grade;
- **Project: 50% (10/20)** of the final grade;
- **Exam: 30% (6/20)** of the final grade.

The **Research Work** corresponds to the first project stage and includes the **Literature Review** and the **Project Proposal**. It is assessed separately from the implementation of the project.

The **Project** component comprises the practical development of the work throughout the semester, including the agreed checkpoints, implementation, experimentation, final report, and final presentation.

Evaluation of the Research Work and Project will focus on, but not be limited to, the following aspects:

- Clear formulation and motivation of the **research problem**.
- Quality and relevance of the **reviewed scientific literature**, including meaningful comparison and connection to the proposed work.
- Meeting the agreed **checkpoints** and **goals**.
- Clarity and quality of the produced **documentation**.
- Proper application of **course-related concepts**.
- **Technical depth and appropriateness** of the proposed solution.
- **Quality of the experimental design**, including meaningful baselines, evaluation criteria, error analysis, uncertainty analysis, sensitivity analysis, or ablation studies whenever appropriate.
- **Novelty** or originality of the proposed investigation, comparison, or system design.

- **Quality and meaningfulness** of results, discussion, and conclusions.
- Quality of the **final presentation** and discussion.

Further remarks:

- **Missing a deadline** will result in a **penalty**. This includes the submission of the Literature Review, Project Proposal, Final Report, and checkpoints.
- The **Final Presentation is mandatory**.
- Even though the project is to be developed in a group, the final grade may be **different for elements of the same group**, depending on individual performance.
- The use of generative AI tools is allowed when appropriate, but students remain responsible for the correctness, understanding, attribution, and reproducibility of all submitted material and should disclose substantial use of such tools.

4 Deadlines

The project has **two** main (hard) deadlines, with results submitted to *Inforestudante*:

- [**31**st **October 2026**] Literature Review & Project Proposal
- [**12**th **December 2026**] Final Report

Two dates must be set by the group in the Project Proposal for checkpoints, where progress will be presented to and discussed with the Professor. These can be selected among the following dates,

- November 9, 12, 16, 19, 23, 26
- December 3, 7

Mark your checkpoint dates at the following form:

- **PL1:** `https://forms.gle/14xy6fqr2jrVnzPC9`

Both members of the group **must** be present in all checkpoints.

Oral presentations should coincide with the final classes, but additional slots may be necessary to accommodate for all groups:

- 14th **December 2026**, 9AM–1PM
- 16th **December 2026**, 9AM–11AM
- 17th **December 2026**, 9AM–11AM

A Suggested Topics

As long as the main requirements are met, the group is free to choose the topic and goals for their project, which should, nevertheless, be discussed with and validated by the Professor. Here are some generic suggestions of possible topics:

- Knowledge Graph Enrichment
  - Select any existing Knowledge Graph
  - Enrich it with entities and / or triples extracted from textual documents [1] or from LLMs
  - Analyse the quality of the extracted knowledge
- Knowledge Graph Extraction or Ontology Learning from LLM
  - Explore prompt engineering for triple extraction [2]
  - Represent extracted knowledge in RDF
  - Evaluate the results
  - May tackle a related public challenge on Knowledge Extraction and LLMs (e.g., LLMs4OL [3], LM-KBC [4])
- Knowledge Injection in Neural Model / LLM [5]
  - Select an NLP task for which an annotated dataset is available (e.g., Text Classification)
  - Explore a LLM for performing the task (e.g., zero-shot few-shot, fine-tuning)
  - Select one or more Knowledge Graphs for providing additional features to inject in the process
  - Analyse the impact of knowledge injection
- Conversion of any dataset to RDF [6]
  - Select any lexicon or dataset
  - Convert this dataset to a RDF-based format (i.e., define classes, properties, consider reusing an existing vocabulary, etc.)
  - Explore the advantages / disadvantages of this format (e.g., query the dataset with SPARQL, consider linking it to other datasets in the Linked Open Data Cloud [7])
  - Use the resulting dataset in some experiment (e.g., classification, question answering)
- Knowledge Graph Embedding [8]
  - Select an NLP task for which an annotated dataset is available

  - Select a knowledge graph that may be useful for the previous task
  - Embed the selected knowledge graph
  - Analyse the advantages of using the graph directly or its embeddings when performing the task.

- GNNs for Text-Attributed Graphs or Knowledge Graph Reasoning
  - Select a graph with textual node / edge attributes and a task such as node classification, link prediction, retrieval, or question answering.
  - Compare a text-only baseline, a graph-only / GNN baseline, and a hybrid graph–language approach.
  - Analyse where relational structure helps and where it does not; optionally study transfer to a new graph or attribute domain.
  - Recent inspiration: G-Retriever [9] and LLM-BP / inference over text-attributed graphs [10].

- Graph Retrieval / Graph-Constrained Reasoning for LLMs
  - Build or reuse a Knowledge Graph and compare different strategies for retrieving or traversing relevant subgraphs for a language-model task.
  - Measure not only answer accuracy but also faithfulness to retrieved paths / triples, retrieval efficiency, and hallucination or unsupported-claim rate.
  - Recent inspiration: Plan-on-Graph [11] and Graph-Constrained Reasoning [12].

- Probabilistic and Uncertainty-Aware Language Systems
  - Study confidence, calibration, abstention, uncertainty-aware retrieval, or information acquisition in a language / QA / extraction task.
  - Compare at least one simple probabilistic baseline (e.g., Naive Bayes, HMM, calibrated classifier, or probabilistic structured model) with a modern neural / language-model baseline.
  - Evaluate whether uncertainty estimates are useful for an explicit downstream decision (e.g., abstain, ask for more context, query a KB, or defer to a human), rather than reporting confidence alone.
  - Recent inspiration: Graph-based Uncertainty for long-form LMs [13], Adaptive Elicitation [14], and Rethinking Aleatoric and Epistemic Uncertainty [15].

- Probabilistic Structured Reasoning around Autoregressive Language Models
  - Combine a simple autoregressive LM with an interpretable structured probabilistic model (e.g., HMM, finite-state model, or another tractable model) to represent global constraints or latent states.
  - Compare local next-token decisions with structured / look-ahead inference and analyse the trade-off between tractability, controllability, and language quality.

  - Recent inspiration: TRACE, which distils an HMM from a language model for tractable probabilistic look-ahead and controllable generation [16].

- Uncertain Graph Structure and GNN Reliability
  - Treat graph edges / relations as uncertain rather than fixed and compare deterministic and probabilistic graph models.
  - Study uncertainty under missing, noisy, or distribution-shifted graph structure; optionally use uncertainty to trigger human review, data acquisition, or graph repair.
  - Recent inspiration: Learning Latent Graph Structures and their Uncertainty [17] and Energy-based Epistemic Uncertainty for GNNs [18].

- Knowledge Conflict: Parametric Memory versus Retrieved / Structured Context
  - Construct cases where an LM's parametric answer conflicts with information in a document, RDF graph, or Knowledge Graph.
  - Measure when the model follows memory versus context, and test simple interventions such as source weighting, uncertainty-aware abstention, or explicit conflict detection.
  - Recent inspiration: Taming Knowledge Conflicts in Language Models [19].

The work may also combine features from more than one suggestion.

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